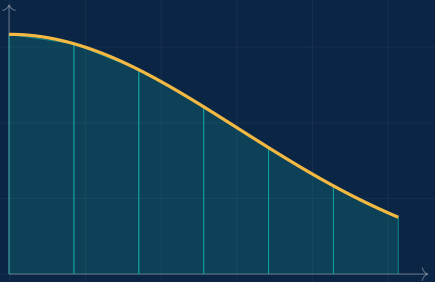
Numerical Integration

From the Trapezoid to Simpson's Rule

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UNSTIM, Benin
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Outline

Why integrate numerically?

The trapezoidal rule

Simpson's rule

Implementation

Experiments

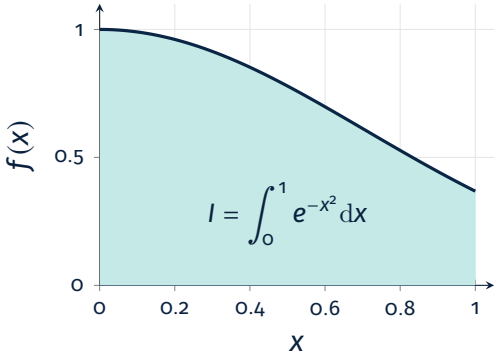
Conclusion

1 Why integrate numerically?

Integrals we cannot solve by hand

- Many integrands have no elementary antiderivative:

$$\int e^{-x^2}\,\mathrm{d}x, \qquad \int \frac{\sin x}{x}\,\mathrm{d}x$$

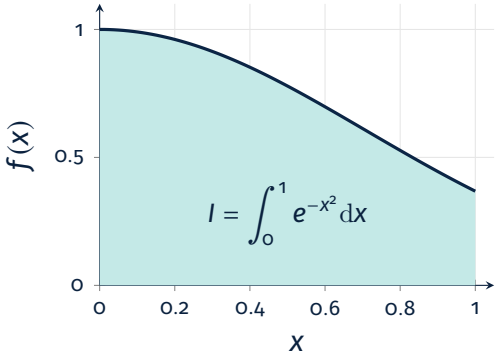


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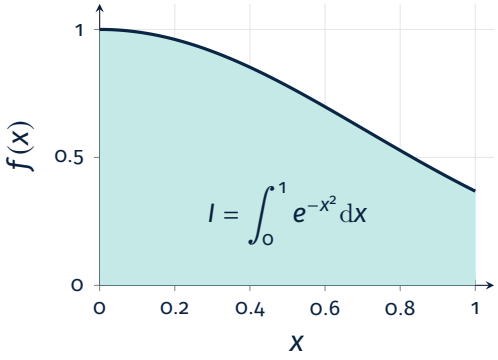


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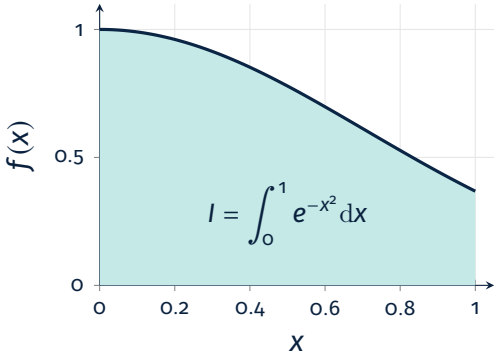


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Our running example

$I = \dfrac{\sqrt{\pi}}{2}\,\mathrm{erf}(1) \approx 0.7468241328$ (exact value, used to measure the errors)

2 The trapezoidal rule

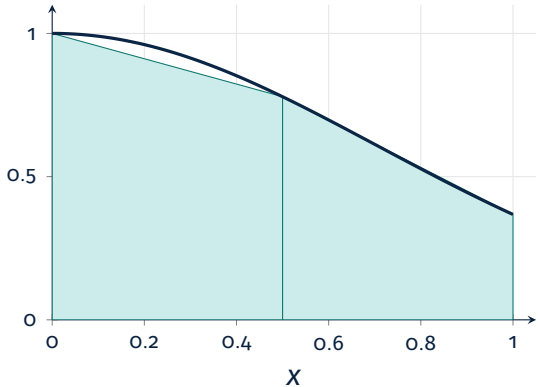
Replace the curve by straight segments

Split $[a, b]$ into n equal pieces of width $h = \dfrac{b-a}{n}$ and approximate f by a line on each piece:

$$T_n = h\left[\tfrac{1}{2}f(a) + \sum_{i=1}^{n-1} f(x_i) + \tfrac{1}{2}f(b)\right]$$

Refine the mesh

$n = 2$ error $\approx 1.5 \cdot 10^{-2}$



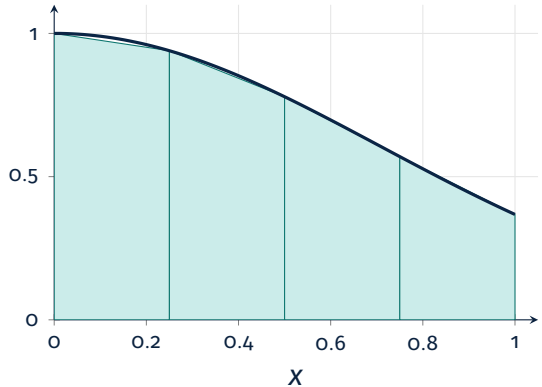
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$n = 4$ error $\approx 3.8 \cdot 10^{-3}$



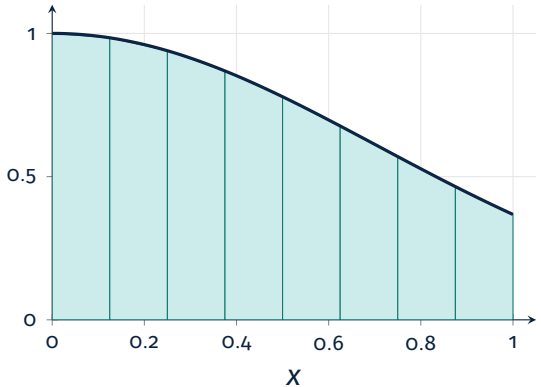
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Refine the mesh

$n = 8$ error $\approx 9.6 \cdot 10^{-4}$



How accurate is it?

Theorem (Trapezoidal error)

If $f \in C^2[a,b]$, *then*

$$|I - T_n| \leq \frac{(b-a)\,h^2}{12} \max_{[a,b]} |f''|.$$

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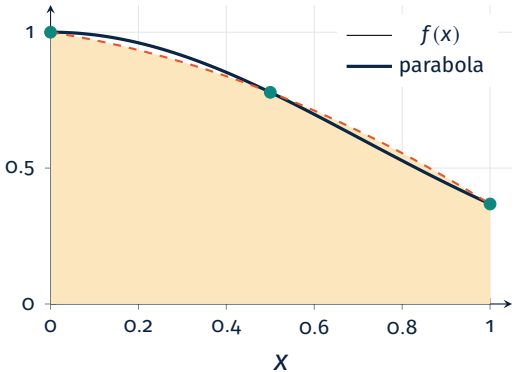
- The error decreases like h^2: doubling n divides the error by about 4.
- The rule is exact for every polynomial of degree ≤ 1.

3 Simpson’s rule

Replace the curve by parabolas

On each pair of pieces, fit a parabola through three points. For $n = 2$:

$$S_2 = \frac{h}{3}\left[f(x_0) + 4f(x_1) + f(x_2)\right]$$



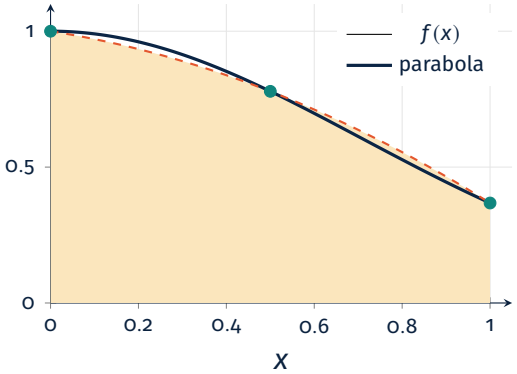
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Check

$h = \frac{1}{2}$ gives
$S_2 = \frac{1}{6}(1 + 4e^{-1/4} + e^{-1}) \approx 0.74718$



The composite rule and its error

For an even number n of pieces:

$$S_n = \frac{h}{3}\left[f(x_0) + 4\sum_{i \text{ odd}} f(x_i) + 2 \sum_{\substack{i \text{ even} \\ 2 \le i \le n-2}} f(x_i) + f(x_n)\right]$$

Theorem (Simpson error)

If $f \in C^4[a,b]$, *then* $|I - S_n| \le \dfrac{(b-a)\,h^4}{180} \max_{[a,b]} |f^{(4)}|$.

The composite rule and its error

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Theorem (Simpson error)

If $f \in C^4[a,b]$, *then* $|I - S_n| \le \dfrac{(b-a)\,h^4}{180} \max_{[a,b]} |f^{(4)}|$.

Consequence

The error decreases like h^4: **doubling** n **divides the error by about** 16**.** Simpson's rule is exact for polynomials up to degree 3.

4 Implementation

The algorithm

Input: function f, interval $[a, b]$, even integer n
Output: approximation S_n of $\int_a^b f$

1: $h \leftarrow (b - a)/n$
2: $s \leftarrow f(a) + f(b)$
3: **for** $i = 1$ **to** $n - 1$ **do**
4: **if** i is odd **then**
5: $s \leftarrow s + 4f(a + ih)$
6: **else**
7: $s \leftarrow s + 2f(a + ih)$
8: **end if**
9: **end for**
10: **return** $h\,s/3$

Cost

$n + 1$ evaluations of f and $O(n)$ operations.

Tip

Choose n even, and double it until two successive results agree.

The same algorithm in Python

```python
import math

def simpson(f, a, b, n):
    if n % 2:
        raise ValueError("n must be even")
    h = (b - a) / n
    s = f(a) + f(b)
    s += 4 * sum(f(a + i*h) for i in range(1, n, 2))
    s += 2 * sum(f(a + i*h) for i in range(2, n-1, 2))
    return s * h / 3

print(simpson(lambda x: math.exp(-x*x), 0, 1, 16))
# 0.7468242574357303
```

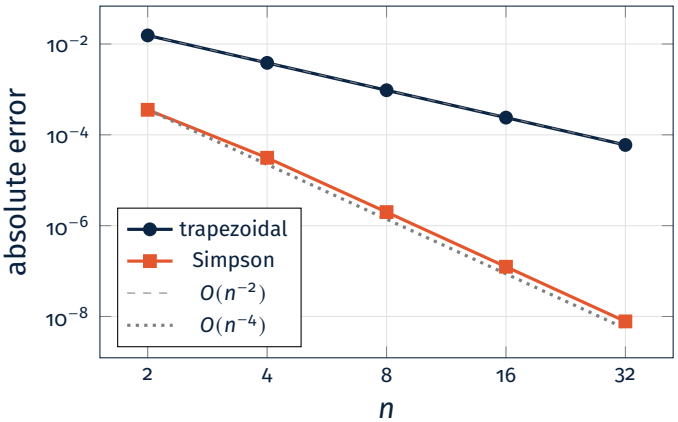
5 Experiments

Numerical results

n	T_n	S_n	$\lvert I - T_n \rvert$	$\lvert I - S_n \rvert$
2	0.731370	0.747180	$1.545 \cdot 10^{-2}$	$3.563 \cdot 10^{-4}$
4	0.742984	0.746855	$3.840 \cdot 10^{-3}$	$3.125 \cdot 10^{-5}$
8	0.745866	0.746826	$9.585 \cdot 10^{-4}$	$1.988 \cdot 10^{-6}$
16	0.746585	0.746824	$2.395 \cdot 10^{-4}$	$1.246 \cdot 10^{-7}$

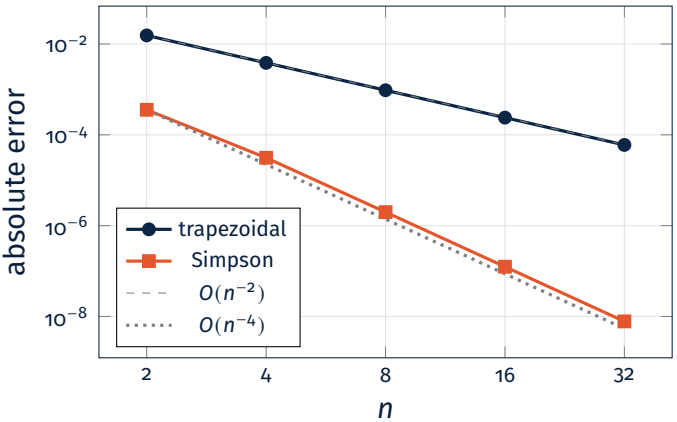
Exact value: $I \approx 0.7468241328$. With only $n = 16$, Simpson's rule is accurate to seven digits.

Convergence study



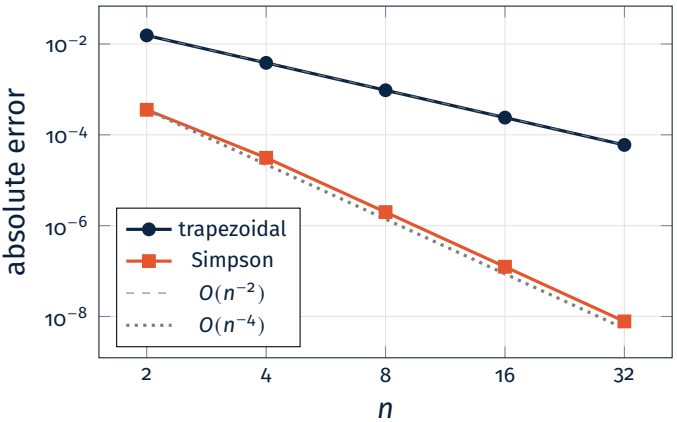
- Trapezoidal: slope -2.

Convergence study



- Trapezoidal: slope -2.
- Simpson: slope -4.

Convergence study



- Trapezoidal: slope -2.
- Simpson: slope -4.
- The measured slopes match the theory.

Which rule should I use?

Trapezoidal rule

- very simple
- works with any n
- order h^2
- good for periodic integrands

Simpson's rule

- same cost per evaluation
- needs an even n
- order h^4
- best for smooth integrands

6 Conclusion

Take-home messages

1. Numerical integration replaces f by simple pieces: lines or parabolas.

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Take-home messages

1. Numerical integration replaces f by simple pieces: lines or parabolas.
2. Error orders: $O(h^2)$ for the trapezoid, $O(h^4)$ for Simpson.
3. On $\int_0^1 e^{-x^2}\,\mathrm{d}x$, $n = 16$ gives $2 \cdot 10^{-4}$ versus $1 \cdot 10^{-7}$.
4. Always check convergence by doubling n.

References I

[1] Richard L. Burden, J. Douglas Faires, and Annette M. Burden. *Numerical Analysis*. 10th ed. Boston: Cengage Learning, 2016.

[2] Kendall E. Atkinson. *An Introduction to Numerical Analysis*. 2nd ed. New York: John Wiley & Sons, 1989.

[3] William H. Press et al. *Numerical Recipes: The Art of Scientific Computing*. 3rd ed. Cambridge: Cambridge University Press, 2007.

Thank you

Questions?

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